SnapLearn AI

- Private Offline Study Companion for Snapdragon-powered HP PCs
- Snapdragon® AI Lab Build & Present Challenge 2026
- One-line pitch: Ask, revise and learn from your own notes without sending them to the cloud.

The Problem

- Cloud study assistants depend on an internet connection.
- Students may hesitate to upload personal or institutional notes to remote services.
- A useful local assistant should work with the student's own material and remain responsive offline.

Our Solution

- Load notes locally.
- Ask questions in natural language.
- Retrieve relevant passages and produce a grounded response.
- Generate revision flashcards from the same material.
- Keep the workflow local-first and privacy-oriented.

Snapdragon Architecture

- Notes → local parsing → Qualcomm AI Hub embedding model → local vector index.
- Question → retrieval → Qualcomm AI Hub small language model → grounded answer.
- Optional voice input can use a compatible local speech model.
- QNN / Hexagon NPU is the intended production execution path on supported Snapdragon HP PCs.

What the Prototype Proves

- A runnable Python reference app is included.
- It works without a cloud API or API key.
- It demonstrates local question answering from loaded notes and flashcard generation.
- The reference implementation does not fabricate Snapdragon benchmark numbers.

Why On-Device AI

- Privacy: notes can remain on the PC.
- Availability: useful without a network connection.
- Cost: no per-question cloud API requirement in the local mode.
- Efficiency: the intended Snapdragon NPU path can be benchmarked for latency and energy.

Roadmap & Validation

- Integrate quantized Qualcomm AI Hub models.
- Enable QNN/Hexagon NPU execution on supported hardware.
- Add PDF/OCR ingestion, voice questions, quizzes and weak-topic tracking.
- Benchmark CPU vs NPU latency, memory, energy and answer quality on real Snapdragon HP hardware.

Demo & Submission

• Run: python app.py

• Use Sample Notes → ask: Why is the domain a subset of R^2?

• Generate Flashcards.

• Files included: README, submission summary, architecture diagram, pitch deck and source code.

• Review eligibility before submitting; the challenge states that participants should own a Snapdragon-powered laptop.